

Substantive Citation Report

Eight articles selected by conference venue and/or citation count, documenting methodological adaptation, technical exposition, comparison and quantizer-design connections.

8

selected articles

36

marked discussion pages

29

countries / territories in
the wider survey

Selection and source documentation

This report includes articles published in conferences rated ICORE2026 A* and/or carrying at least 100 Google Scholar citations in the saved snapshot. Counts belong to the citing articles' Scholar records or version clusters. Each entry identifies its reviewed publication version and the contribution supported by the source passage.

Authorship links and documented scholarly use

Each article identifies the cited work and authors, the specific research connection and the original discussion pages. The bibliography link establishes authorship of the corresponding work by Sambhav R. Jain and coauthors.

Orange rectangles mark selected discussion, citations and TQT results; blue rectangles identify the corresponding bibliography entries. Pages 1–36 of the marked packet follow the article order in this report and retain original page numbers. Article numbers identify the corresponding marked files.

International citation survey. The wider survey maps 146 citing works to 157 institution/location entries across 29 countries/territories. These totals describe the broader citation network, separately from this report's selection.

[Report PDF](#) · [Marked discussions](#) · [pages 1–36](#) · [Interactive map](#)

Snapshot: October 7, 2026. [Sambhav Jain's Google Scholar profile](#). Counts belong to the saved citing-article records or version clusters.

Eight substantive citing articles

No.	Article / venue	Scholar	Selection basis
01	Understanding and Overcoming the Challenges of Efficient Transformer Quantization EMNLP 2021	271	ICORE2026 A* venue + ≥ 100 Scholar citations
02	Up or Down? Adaptive Rounding for Post-Training Quantization ICML 2020	1,224	ICORE2026 A* venue + ≥ 100 Scholar citations
03	A White Paper on Neural Network Quantization arXiv / Qualcomm white paper, 2021	1,362	≥ 100 Scholar citations
04	Mixed Precision DNNs: All you need is a good parametrization ICLR 2020	230	ICORE2026 A* venue + ≥ 100 Scholar citations
05	Bayesian Bits: Unifying Quantization and Pruning NeurIPS 2020	207	ICORE2026 A* venue + ≥ 100 Scholar citations
06	Bringing AI to edge: From deep learning's perspective Neurocomputing 485 (2022)	247	≥ 100 Scholar citations
07	HMQ: Hardware Friendly Mixed Precision Quantization Block for CNNs ECCV 2020	95	ICORE2026 A* venue
08	QKD: Quantization-aware Knowledge Distillation arXiv preprint, 2019	125	≥ 100 Scholar citations

SUBSTANTIVE CITATIONS

ARTICLE 01 · SUBSTANTIVE CITATION REPORT

Understanding and Overcoming the Challenges of Efficient Transformer Quantization

Yelysei Bondarenko; Markus Nagel; Tijmen Blankevoort
EMNLP 2021 · 271 Google Scholar citations

ICORE2026 A* venue + ≥ 100 Scholar citations

Documented contribution. Qualcomm researchers explicitly adapt the jointly credited LSQ/TQT procedure from Esser et al. and Jain et al. to learn weight and activation ranges for BERT-like transformer models.

Methodological adaptation. Adapts trainable weight and activation ranges to BERT-like models, jointly crediting the procedures of Jain et al. and Esser et al.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. [Bibliography p. 11](#).

Reviewed version: Published proceedings PDF. **Pinpoint:** Physical PDF pp. 3, 7, learnable weight/activation ranges and their adaptation to transformers; bibliography p. 11 / printed p. 7957. Source E-0046; marked copy E-0884.

Count: E-0430, data.organic_results[6].inline_links.cited_by.total. Venue: E-0326.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 01-Transformer-Quantization-MARKED.pdf

ARTICLE 02 · SUBSTANTIVE CITATION REPORT

Up or Down? Adaptive Rounding for Post-Training Quantization

Markus Nagel; Rana Ali Amjad; Mart van Baalen; Christos Louizos; Tijmen Blankevoort
ICML 2020 · 1,224 Google Scholar citations

ICORE2026 A* venue + ≥ 100 Scholar citations

Documented contribution. The AdaRound paper names Jain et al., alongside Esser et al., when explaining how quantization ranges can be learned during training instead of being set manually.

Named technical recognition. Explains the contribution of learning quantization ranges in the related-work analysis of AdaRound, with joint credit to Jain et al. and Esser et al.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. [Bibliography p. 10](#).

Reviewed version: Published PMLR PDF; physical pages retained. **Pinpoint:** Physical PDF p. 5, section 4 Background and related work; bibliography physical p. 10. Source E-0321; marked copy E-0885.

Count: E-0430, data.organic_results[2].inline_links.cited_by.total. Venue: E-0329.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 02-AdaRound-MARKED.pdf

SUBSTANTIVE CITATIONS

ARTICLE 03 · SUBSTANTIVE CITATION REPORT

A White Paper on Neural Network Quantization

Markus Nagel; Marios Fournarakis; Rana Ali Amjad; Yelysei Bondarenko; Mart van Baalen; Tijmen Blankevoort
arXiv / Qualcomm white paper, 2021 · 1,362 Google Scholar citations

≥100 Scholar citations

Documented contribution. The quantization white paper explains learning quantization parameters through the straight-through estimator, credits Jain et al. alongside Esser et al. and Bhalgat et al., and develops the corresponding gradient derivation.

Technical exposition. Develops the gradient calculation for learnable quantization parameters after crediting Jain et al., Esser et al. and Bhalgat et al.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. [Bibliography p. 26.](#)

Reviewed version: Reviewed arXiv white paper. **Pinpoint:** Physical PDF pp. 20, 22, gradient derivation and QAT initialization; bibliography p. 26. Source E-0090; marked copy E-0886.

Count: E-0430, `data.organic_results[1].inline_links.cited_by.total`.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 03-White-Paper-MARKED.pdf

ARTICLE 04 · SUBSTANTIVE CITATION REPORT

Mixed Precision DNNs: All you need is a good parametrization

Stefan Uhlich; Lukas Mauch; Fabien Cardinaux; Kazuki Yoshiyama; Javier Alonso García; Stephen Tiedemann; Thomas Kemp; Akira Nakamura
ICLR 2020 · 230 Google Scholar citations

ICORE2026 A* venue + ≥100 Scholar citations

Documented contribution. Sony researchers describe TQT's trained dynamic range and train a TQT-style fixed-bitwidth model for their experimental comparison with mixed-precision quantization.

Implemented experimental comparison. Trains a TQT-style model as a fixed-bitwidth benchmark for the authors' concurrently developed mixed-precision method.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. [Bibliography p. 10.](#)

Reviewed version: Author-hosted arXiv copy bearing ICLR 2020 publication header. **Pinpoint:** PDF/printed pp. 1–2, 4–5, 8–9, introduction, quantizer constraints, experimental setup and TQT result rows; bibliography p. 10. Source E-0087; marked copy E-0887.

Count: E-0430, `data.organic_results[11].inline_links.cited_by.total`. Venue: E-0328.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 04-Mixed-Precision-DNNs-MARKED.pdf

SUBSTANTIVE CITATIONS

ARTICLE 05 · SUBSTANTIVE CITATION REPORT

Bayesian Bits: Unifying Quantization and Pruning

Mart van Baalen; Christos Louizos; Markus Nagel; Rana Ali Amjad; Ying Wang; Tijmen Blankevoort; Max Welling
NeurIPS 2020 · 207 Google Scholar citations

ICORE2026 A* venue + ≥ 100 Scholar citations

Documented contribution. The Bayesian Bits authors credit two papers with independently introducing joint learning of scale and model parameters. Reference [15] identifies the TQT work by Jain, Gural, Wu and Dick.

Attribution of a research contribution. Credits reference [15] as one of two papers that independently introduced joint learning of quantization scale and network parameters.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. [Bibliography p. 10.](#)

Reviewed version: Published NeurIPS PDF. **Pinpoint:** PDF/printed pp. 6, 8, scale learning and TQT in Figure 2b and its caption; bibliography p. 10, reference [15]. Source E-0324; marked copy E-0888.

Count: E-0430, `data.organic_results[9].inline_links.cited_by.total`. Venue: E-0331.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 05-Bayesian-Bits-MARKED.pdf

ARTICLE 06 · SUBSTANTIVE CITATION REPORT

Bringing AI to edge: From deep learning's perspective

Di Liu; Hao Kong; Xiangzhong Luo; Weichen Liu; Ravi Subramaniam
Neurocomputing 485 (2022) · 247 Google Scholar citations

≥ 100 Scholar citations

Documented contribution. The edge-AI survey gives Jain et al. a dedicated methodological description of trained uniform quantization for accurate and efficient neural-network inference on fixed-point hardware.

Named survey discussion. Explains trained uniform quantization for fixed-point inference in a survey of deep learning at the edge; reference [121] identifies Jain et al.'s paper.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. [Bibliography p. 20.](#)

Reviewed version: arXiv author manuscript; journal publication verified separately. **Pinpoint:** PDF/printed pp. 8–9, TQT comparison-table row and named quantization discussion; bibliography p. 20, reference [121]; affiliations p. 1. Source E-0044; marked copy E-0889.

Count: E-0366, `data.organic_results[0].inline_links.cited_by.total`.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 06-Bringing-AI-to-Edge-MARKED.pdf

SUBSTANTIVE CITATIONS

ARTICLE 07 · SUBSTANTIVE CITATION REPORT

HMQ: Hardware Friendly Mixed Precision Quantization Block for CNNs

Hai Victor Habi; Roy H. Jennings; Arnon Netzer

ECCV 2020 · 95 Google Scholar citations

ICORE2026 A* venue

Documented contribution. HMQ states that the quantizer it uses is similar to the quantizer of Jain et al. identified by reference [24], then presents its hardware-friendly mixed-precision formulation.

Quantizer-design connection. Identifies the quantizer in reference [24] as similar to the one used in HMQ, followed by signed and unsigned quantizer definitions in equations 2 and 3.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 17](#).

Reviewed version: Author arXiv manuscript; ECCV publication verified separately. **Pinpoint:** Physical PDF pp. 1–4, hardware-friendly quantization citations and quantizer definition; bibliography p. 17. Source E-0043; marked copy E-0891.

Count: E-0368, `data.organic_results[0].inline_links.cited_by.total`. Venue: E-0585.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 07-HMQ-MARKED.pdf

ARTICLE 08 · SUBSTANTIVE CITATION REPORT

QKD: Quantization-aware Knowledge Distillation

Jangho Kim; Yash Bhalgat; Jinwon Lee; Chirag Patel; Nojun Kwak

arXiv preprint, 2019 · 125 Google Scholar citations

≥100 Scholar citations

Documented contribution. QKD describes trainable intervals in LSQ and TQT alongside other trainable quantization approaches, and states that it uses these approaches in its baseline before applying knowledge distillation.

Use in a research implementation. Includes TQT [17] among the jointly credited trainable quantization approaches used in the baseline. The reported results concern quantization combined with knowledge distillation.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 9](#).

Reviewed version: Reviewed arXiv manuscript. **Pinpoint:** Physical PDF pp. 2–3, 7, trainable intervals, dequantization and depthwise-layer sensitivity; bibliography p. 9, reference [17]. Source E-0056; marked copy E-0892.

Count: E-0378, `data.organic_results[0].inline_links.cited_by.total`.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 08-QKD-MARKED.pdf

Citation coverage across the research portfolio

Entry	Cited work	Profile citations	Records retrieved	Grouped works	Mapped works
01	Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks	227	216	206	105
02	Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware	71	71	69	34
03	Degradation of 2DEG transport properties in GaN-capped AlGaIn/GaN heterostructures at 600° C in oxidizing and inert environments	22	22	21	6
04	Interdigitated Pt-GaN Schottky interfaces for high-temperature soot-particulate sensing	15	15	15	1
05	Training a fully convolutional neural network to route integrated Circuits	13	13	13	1
06	A novel self-consistent model based optimal filter design for the improved dynamic performance of 3-phase PLLs for phase tracking under grid imperfections (Part 2)	4	4	4	1
07	System and method for implementing neural networks in integrated circuits	1	1	1	0
08	Gallium Nitride Microelectronics for High-Temperature Environments	1	1	1	1
09	An ML Compilation Stack for Autonomous Vehicles using MLIR	0	0	0	0
10	Systematic DL Optimizations for Accurate and Efficient TensorRT Deployments on Cruise's Self-Driving Fleet	0	0	0	0
11	TQT: Trained Quantization Thresholds	0	0	0	0

A citing work can appear in several publication rows. Unique survey totals group overlapping records and publication versions.

The saved Scholar profile records 354 citations. The inventory retains 323 works from 337 archived Scholar records. Publication affiliations establish locations for 146 works; the remaining 177 have their review status recorded in the inventory.

Dated Scholar responses document discovery and article-level counts. Source PDFs establish the selected discussions and affiliation evidence. The coverage table and inventory provide the denominators for each measure.

Publication-sourced affiliations and research connections

The map presents 157 institution/location entries from 146 grouped citing works in 29 countries/territories. Each pin opens the institutions, citing papers and affiliation-page evidence. Several institutions in one city share a display pin; multi-institution papers contribute several locations.

Affiliations come from the reviewed publication. Explicit campus and city details determine placement; ROR supplies institution-city geography when the publication lists an institution without a city. Hollow markers show country-level affiliations. Coordinates are geographic display anchors.

The map documents the geographic reach of citing authors' publication affiliations, including coauthor- and employer-linked works. Article passages establish methodological adaptation, implementation, comparison, design connections or technical exposition. Joint attribution is retained where several methods are credited.

Research files

[Marked discussions](#): pages 1–36 in this report's article order.

Marked-Articles/: complete articles with original page numbers and source links.

Selected-Articles.csv: selection membership, counts, evidence types and source pages.

Discussion-Page-Index.csv: article-to-packet page mapping.

Sources/: dated Scholar, publication, venue, geography and affiliation captures.

[Substantive Citation Report](#) · [Interactive map](#)

Counts refer to the saved October 7, 2026 snapshot. Hashes and source provenance are recorded in the research manifests.